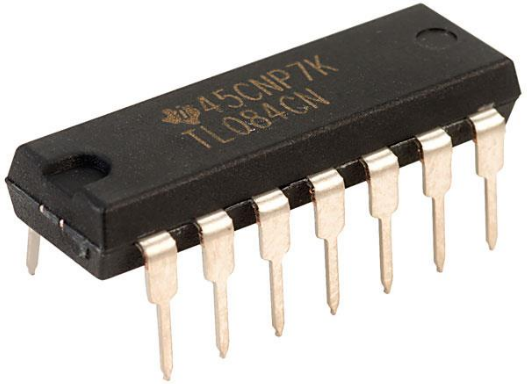
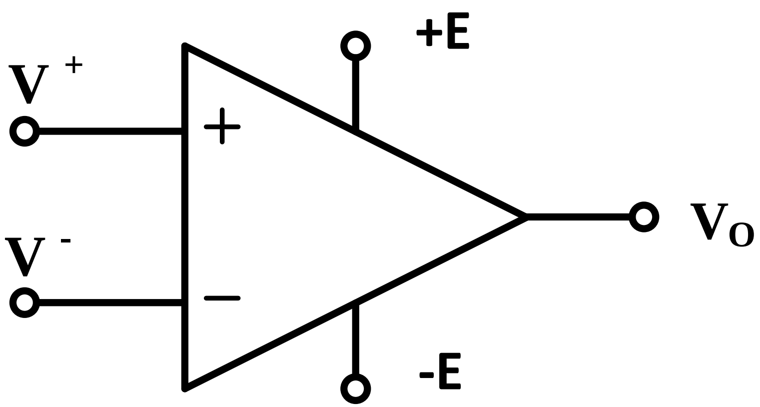
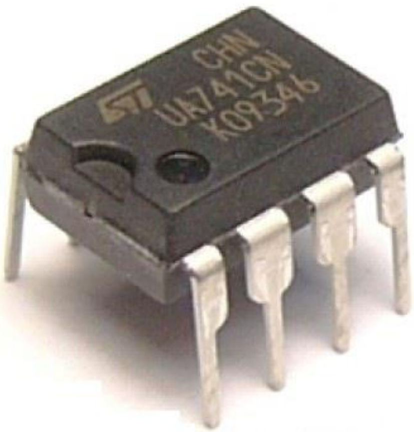
Bazele Circuitelor Electronice

Amplificatoare Operaționale



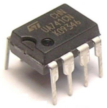
Simbol





$$\boxed{v_o = A_D \cdot \left(v^+ - v^-\right) = A_D \cdot V_D}$$

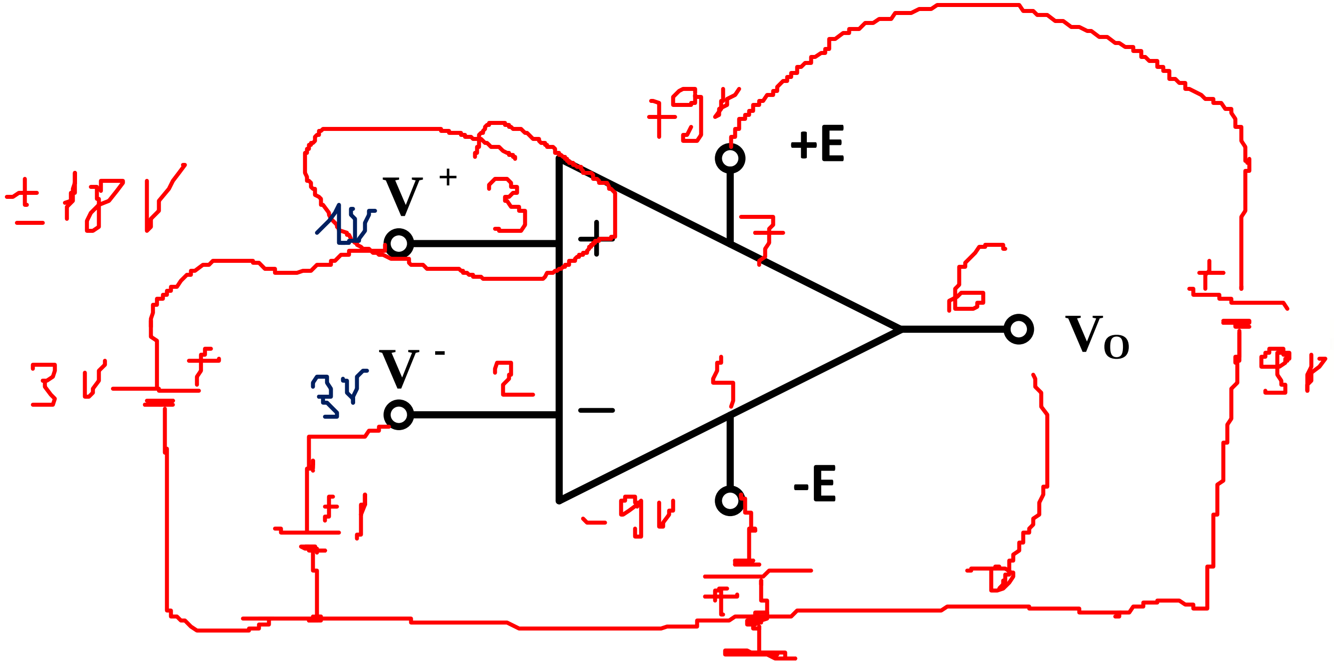
A_D – amplificare diferențială în buclă deschisă

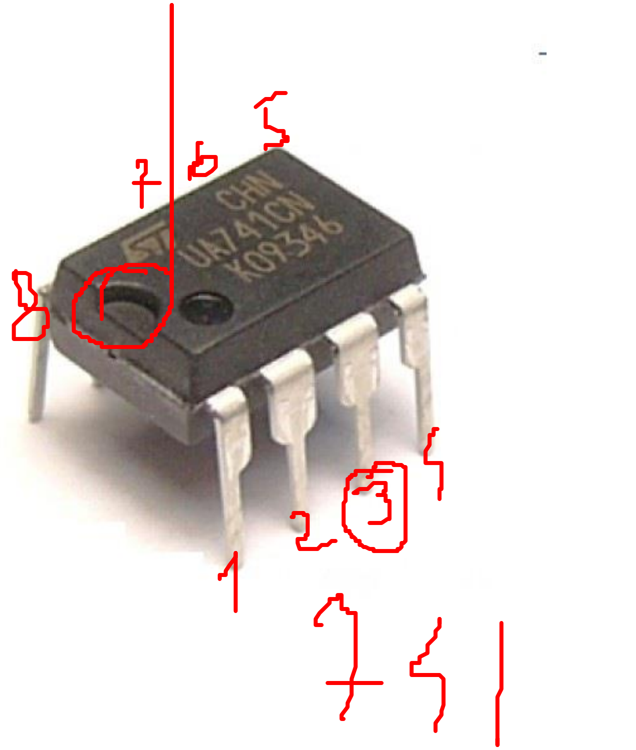


Simbol

DIP 8

$\pm E = \pm 5 \ldots \pm 18\,V$





$7 \; 4 \; 1$

$V_o = \infty(1-3) =$

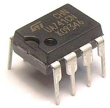
$= \infty(-2) = -\infty$

$= -9$

$$\boxed{v_o = A_D \cdot \left(v^+ - v^-\right) = A_D \cdot V_D}$$

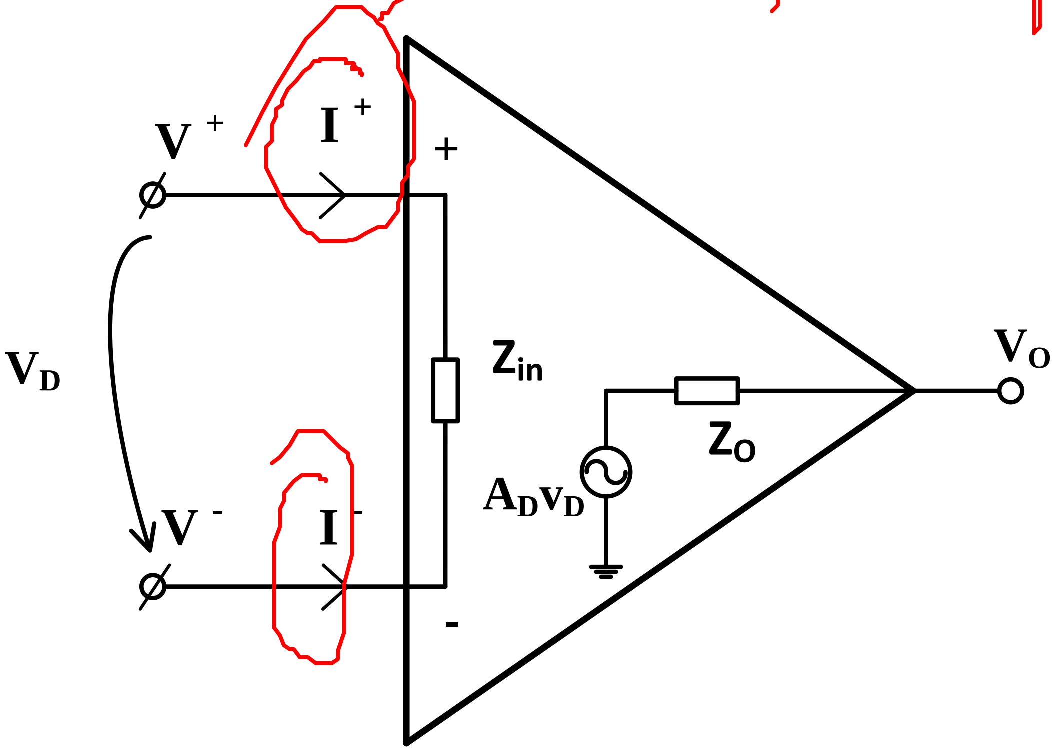
$V_o = A_D(v^+ - v^-) = \infty(3-1) = \infty \cdot 2 = \infty \to +9V$

A_D – amplificare diferențială în buclă deschisă

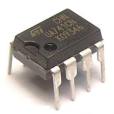


Model electric

→ curenți de polarizare

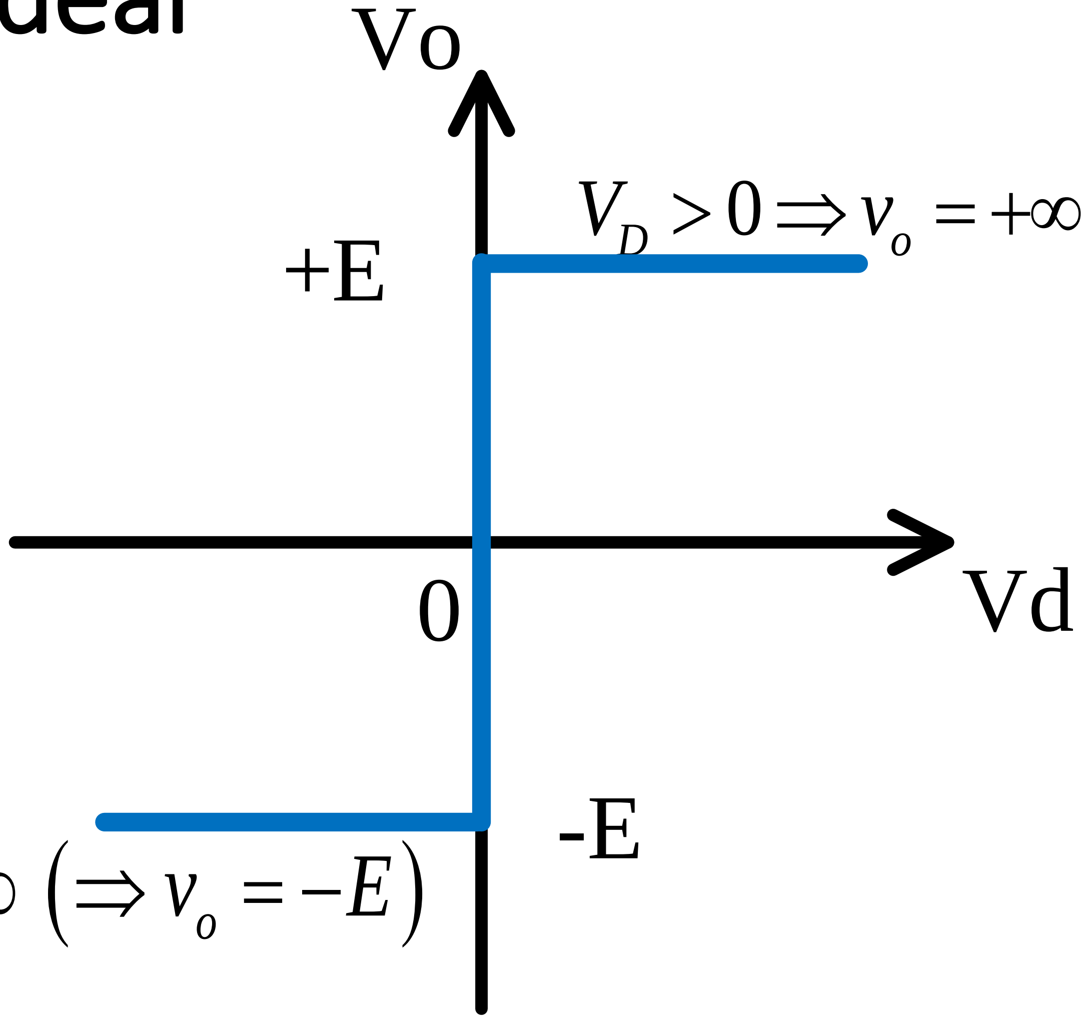


AO – IDEAL	AO – REAL
$A_D = \infty$	$A_D = 10^6..10^{12}$
$Z_i \to \infty$	$Z_i = 10^6..10^{10}\,\Omega$
$i^+ = i^- = 0$	$i^+ \neq i^- \neq 0[\text{nA}; \mu A]$
$V_{off} = 0 \Rightarrow v^+ = v^-$	$v^+ = v^- + V_{off}$
$R_o = 0$	$R_0 \neq 0[\text{zeci, sute}\,\Omega]$
$V_{o\max} = \pm E$	$V_{o\max} = \pm(E - 1...2V)$

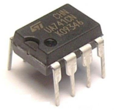


Caracteristica Statică de Transfer în Tensiune (CSTV), **AO ideal**

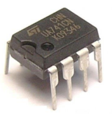
$$V_D > 0 \Rightarrow v_o = +\infty \;\left(\Rightarrow v_o = +E\right)$$



$$V_D < 0 \Rightarrow v_o = -\infty \;\left(\Rightarrow v_o = -E\right)$$

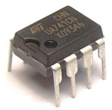


Circuite fundamentale cu AO



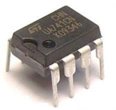
Circuite cu AO

- fără Reacție
  - Comparator neinversor fără Reacție
  - Comparator inversor fără Reacție
- cu Reacție
  - Reacție Pozitivă
    - Comparator neinversor cu Reacție
    - Comparator inversor cu Reacție
  - Reacție Negativă
    - Amplificator neinversor
    - Amplificator inversor

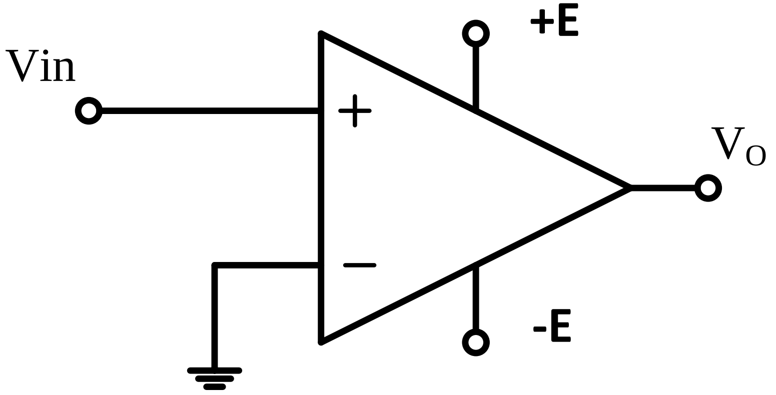


Circuite cu AO

- fără Reacție
  - Comparator neinversor fără Reacție
  - Comparator inversor fără Reacție
- cu Reacție
  - Reacție Pozitivă
    - Comparator neinversor cu Reacție
    - Comparator inversor cu Reacție
  - Reacție Negativă
    - Amplificator neinversor
    - Amplificator inversor



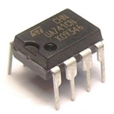
Comparator neinversor fără reacție



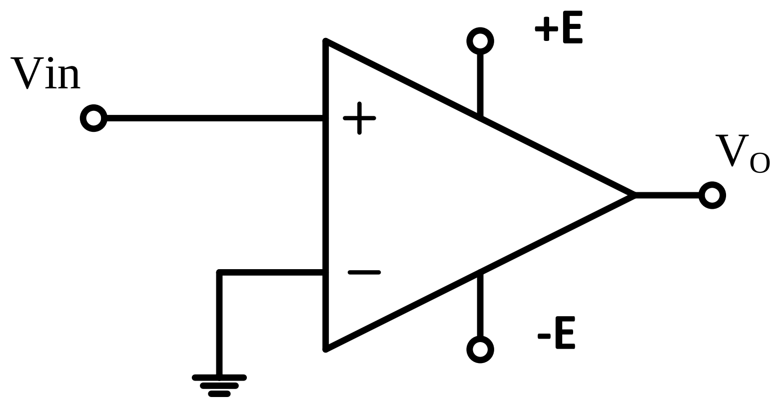
$$v^{+} = v_{in} = V_{\max} \sin(\omega t)$$

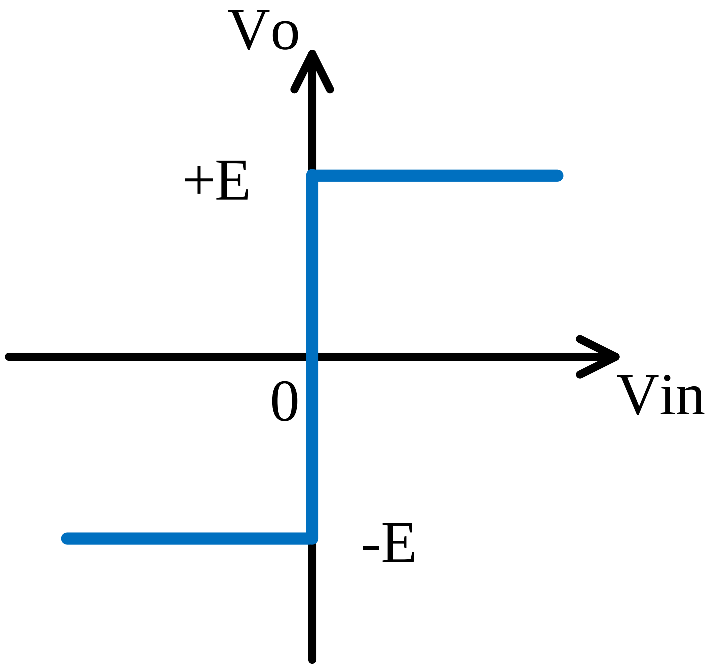
$$v^{-} = 0V$$

$$v_o = A_D \cdot \left(v^{+} - v^{-}\right) = \infty\left(v_{in} - 0\right) = \infty\left[\mathrm{V}_{\max} \sin(\omega t) - 0\right] = \begin{cases} +\infty, v_{in} \geq 0 \\ -\infty, v_{in} < 0 \end{cases}$$

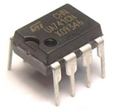


Comparator neinversor fără reacție - CSTV

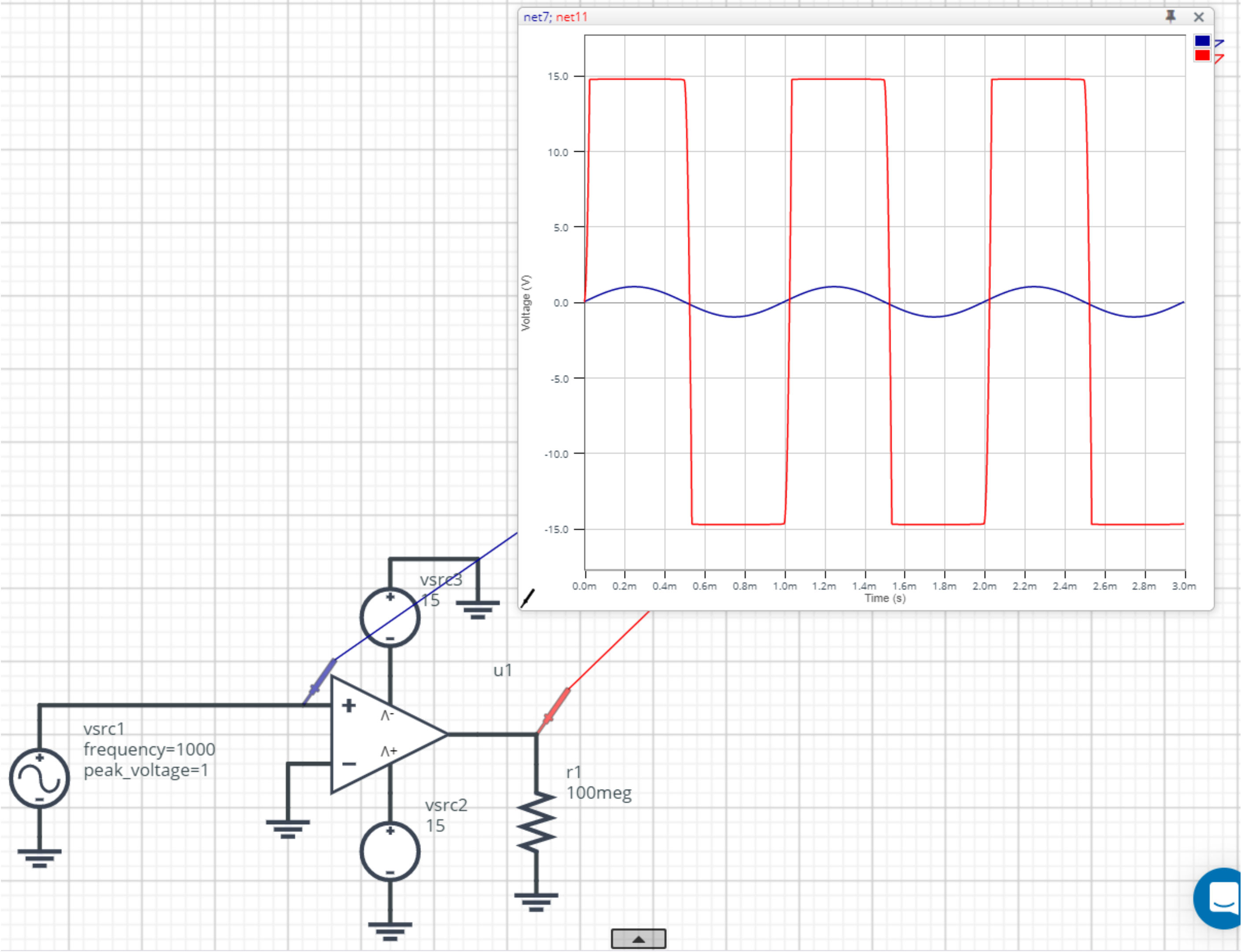




$$v_o = \begin{cases} +E, v_{in} \geq 0 \\ -E, v_{in} < 0 \end{cases}$$



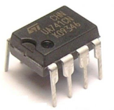
Comparator neinversor fără reacție - CSTV



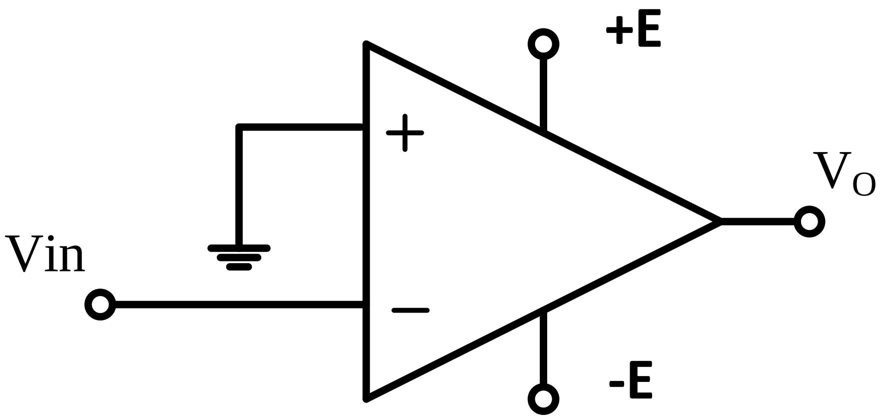
Comparator neinversor fara reactie

Circuite cu AO

- fără Reacție
  - Comparator neinversor fără Reacție
  - Comparator inversor fără Reacție
- cu Reacție
  - Reacție Pozitivă
    - Comparator neinversor cu Reacție
    - Comparator inversor cu Reacție
  - Reacție Negativă
    - Amplificator neinversor
    - Amplificator inversor



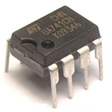
Comparator inversor fără reacție



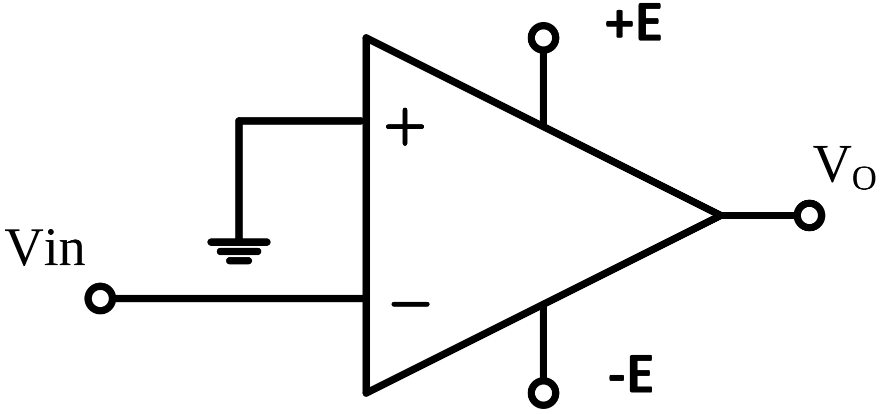
$$v^{+} = 0V$$

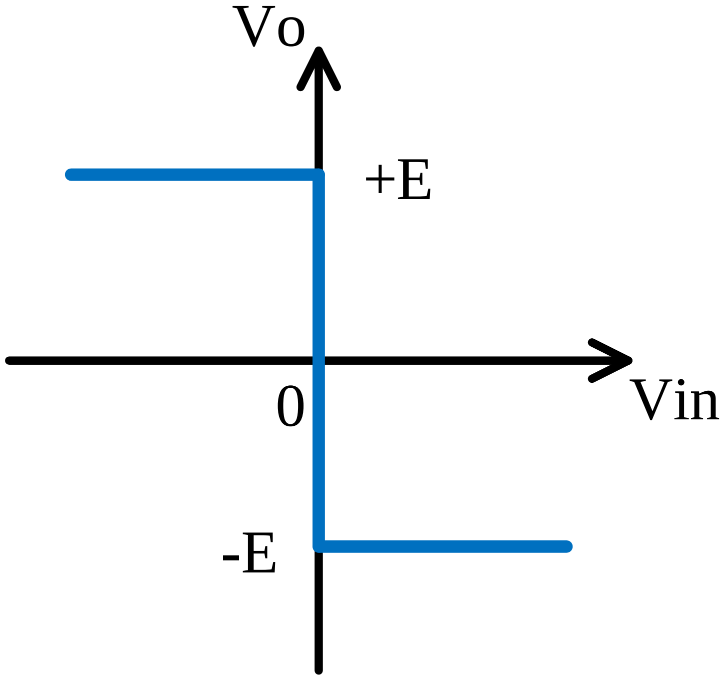
$$v^{-} = v_{in} = V_{\max} \sin\left(\omega t\right)$$

$$v_o = A_D \cdot \left(v^{+} - v^{-}\right) = \infty\left(0 - v_{in}\right) = \infty\left[0 - \mathrm{V}_{\max} \sin\left(\omega t\right)\right] = \begin{cases} -\infty, v_{in} \geq 0 \\ +\infty, v_{in} < 0 \end{cases}$$

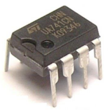


Comparator inversor fără reacție - CSTV

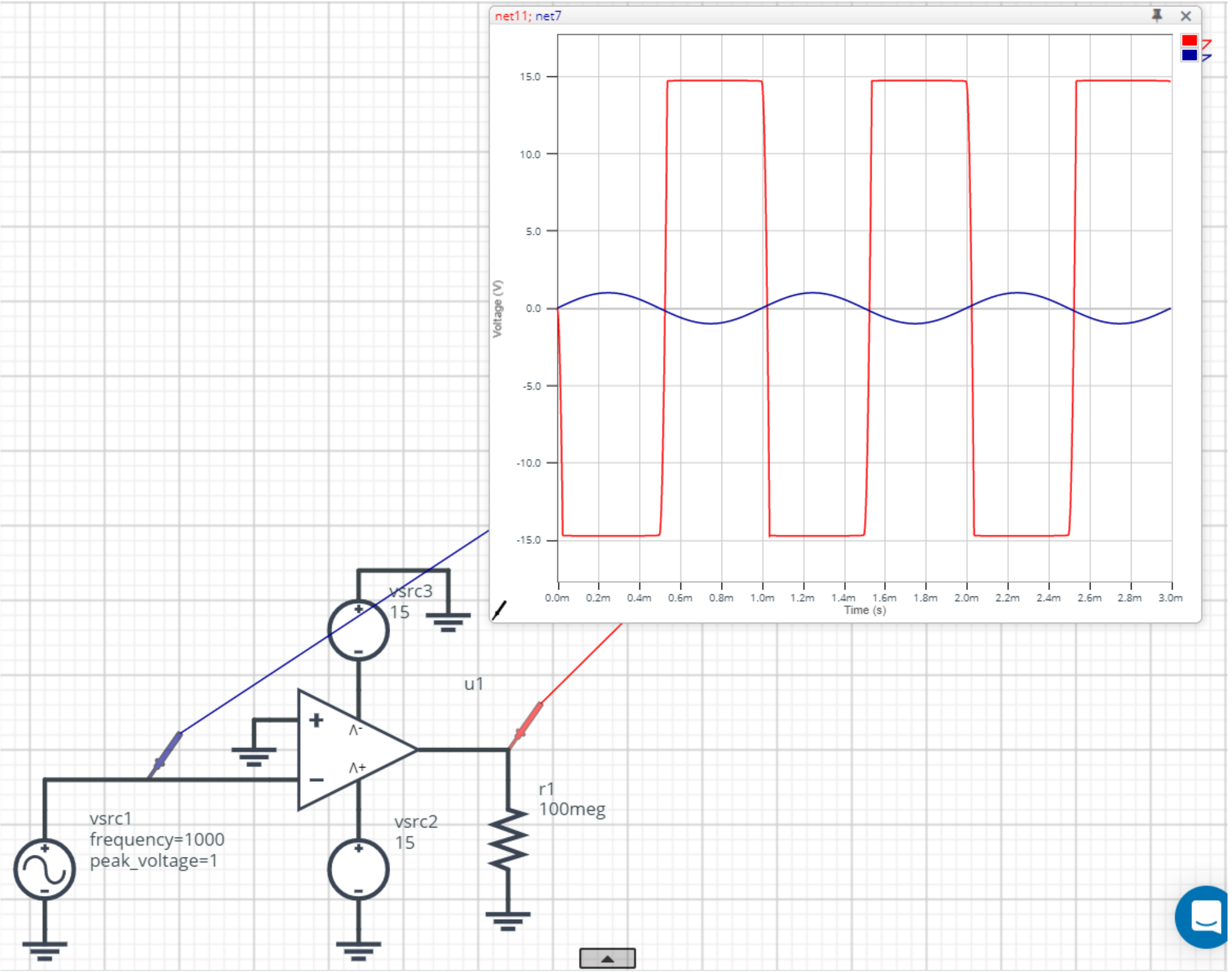




$$v_o = \begin{cases} -E, v_{in} \geq 0 \\ +E, v_{in} < 0 \end{cases}$$



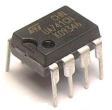
Comparator inversor fără reacție - CSTV



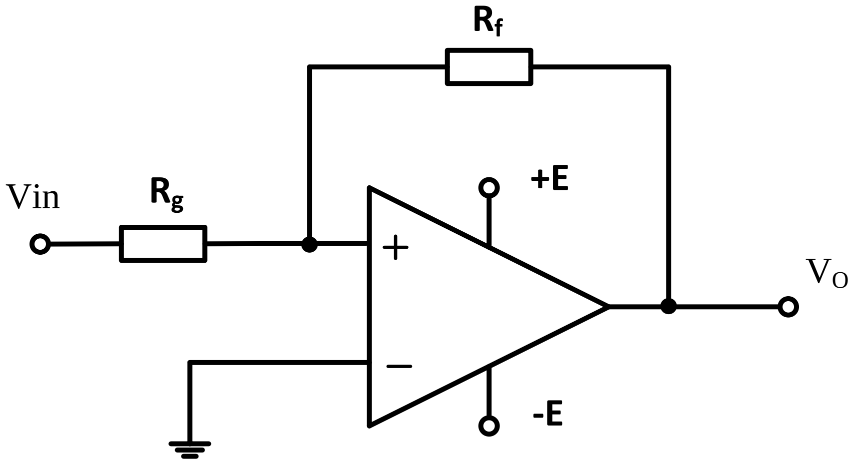
SVC Comparator inversor fara reactie

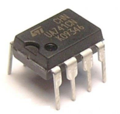
Circuite cu AO

- fără Reacție
  - Comparator neinversor fără Reacție
  - Comparator inversor fără Reacție
- cu Reacție
  - Reacție Pozitivă
    - Comparator neinversor cu Reacție
    - Comparator inversor cu Reacție
  - Reacție Negativă
    - Amplificator neinversor
    - Amplificator inversor

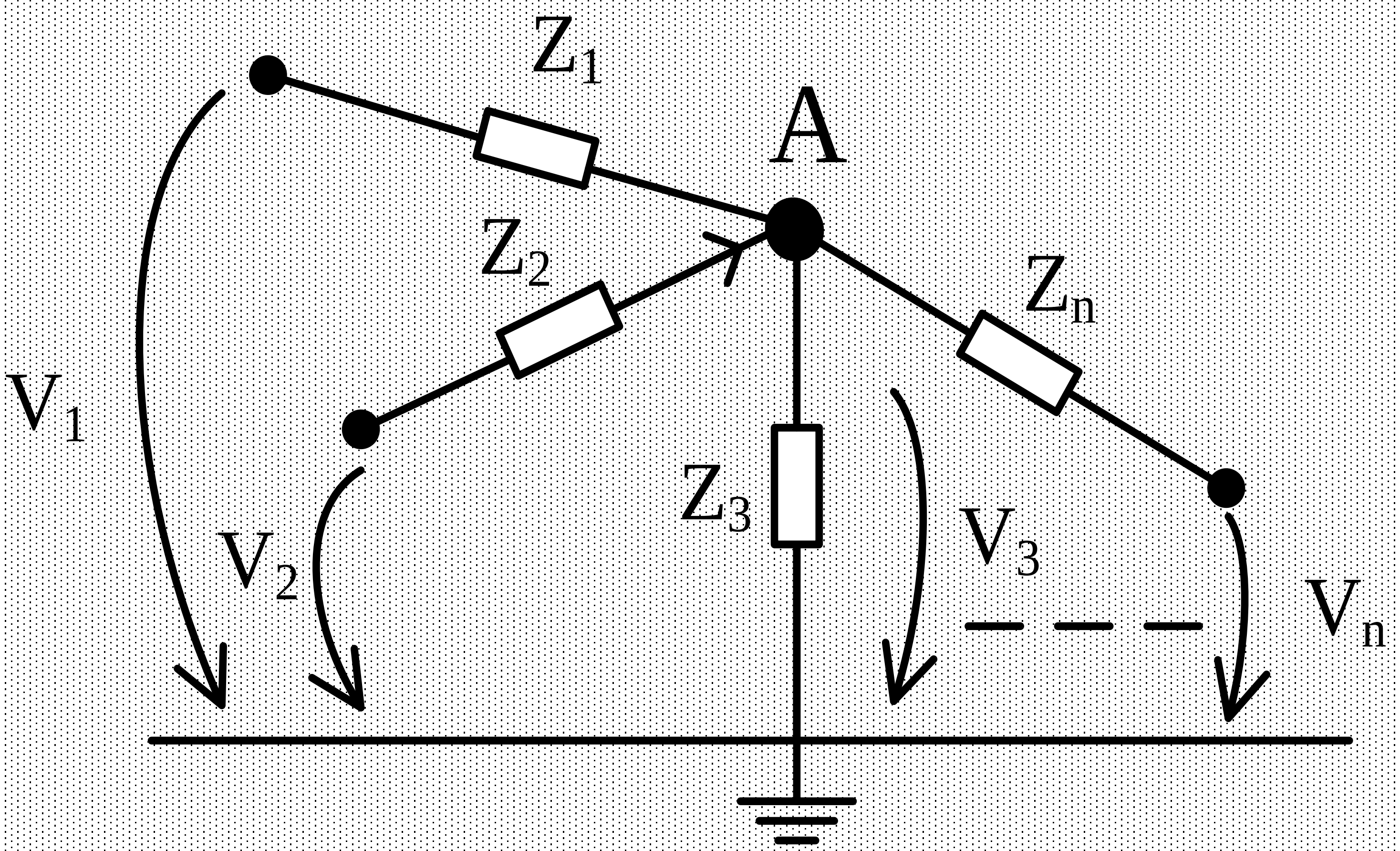


Comparator neinversor cu reacție





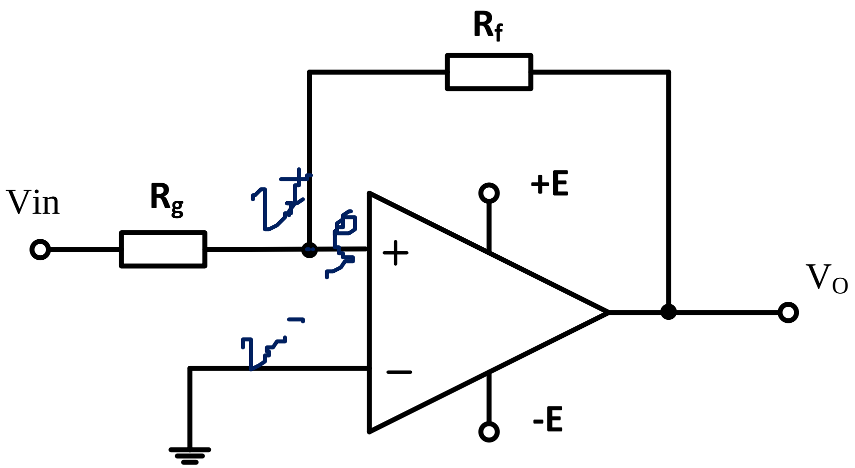
Teorema lui Millman



$$V_A = \frac{\sum_{i=1}^{n} \frac{V_i}{Z_i}}{\sum_{i=1}^{n} \frac{1}{Z_i}}$$

Comparator neinversor cu reacție

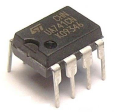
la comutare $v^+ = v^-$



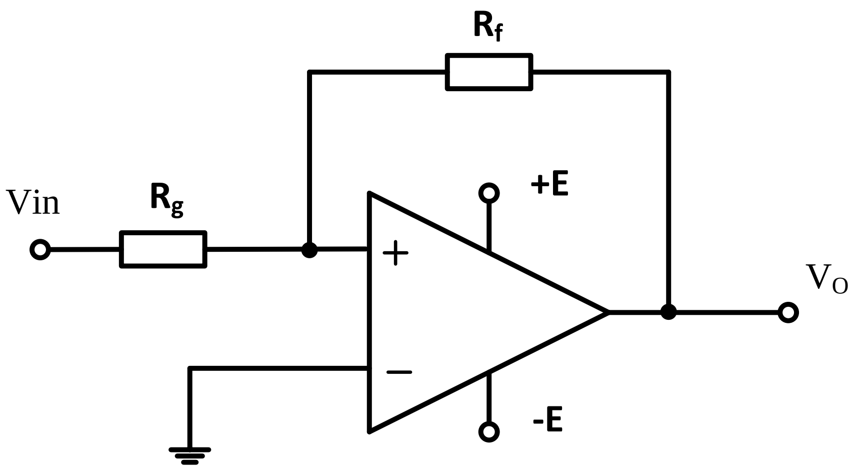
$$\begin{cases} v^- = 0 \\ v^+ = \dfrac{\dfrac{v_o}{R_f} + \dfrac{v_i}{R_g}}{\dfrac{1}{R_f} + \dfrac{1}{R_g}} = 0 \end{cases} \Rightarrow \frac{v_o}{R_f} + \frac{v_i}{R_g} = 0 \Rightarrow \frac{V_{prag}}{R_g} = -\frac{v_o}{R_f}$$

pt. $v_{in} = V_p \Rightarrow v^+ = v^-$

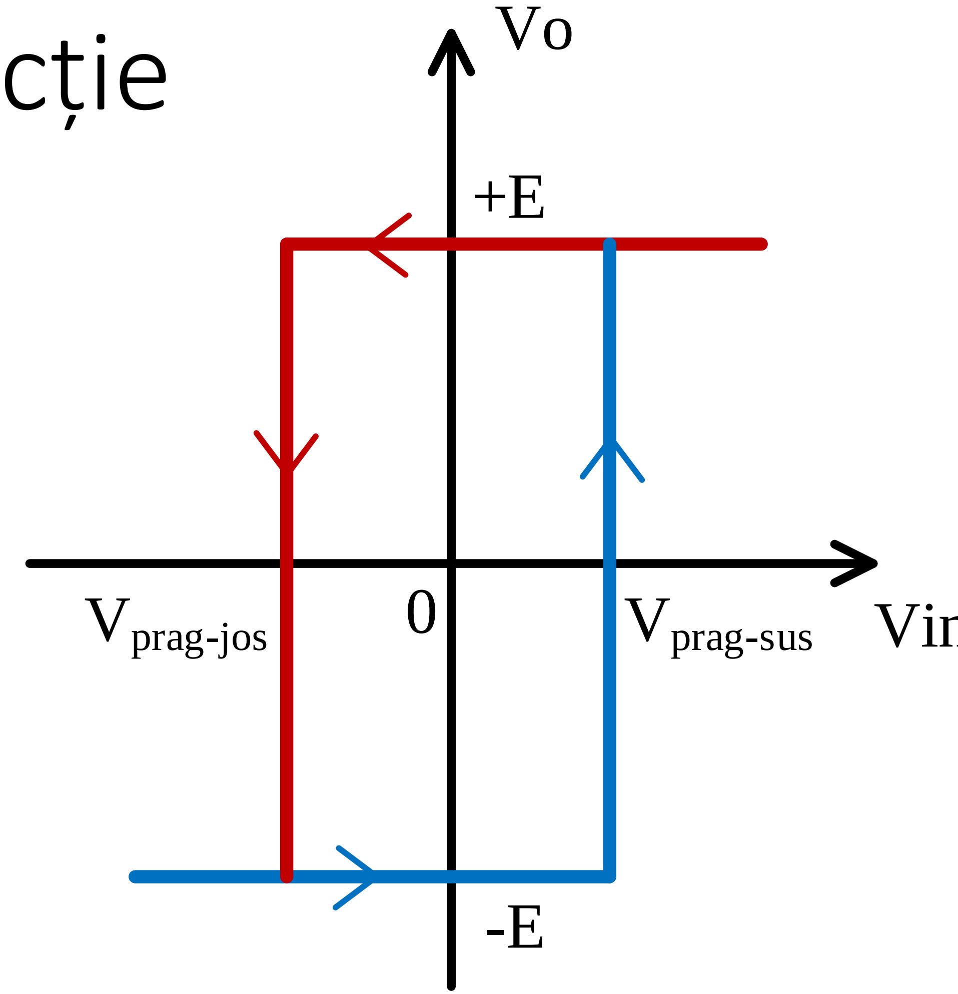
$$\Rightarrow \frac{V_{prag}}{R_g} = -\frac{\pm E}{R_f} \Rightarrow \boxed{V_{prag} = \pm \frac{R_g}{R_f} E}$$

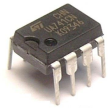


Comparator neinversor cu reacție

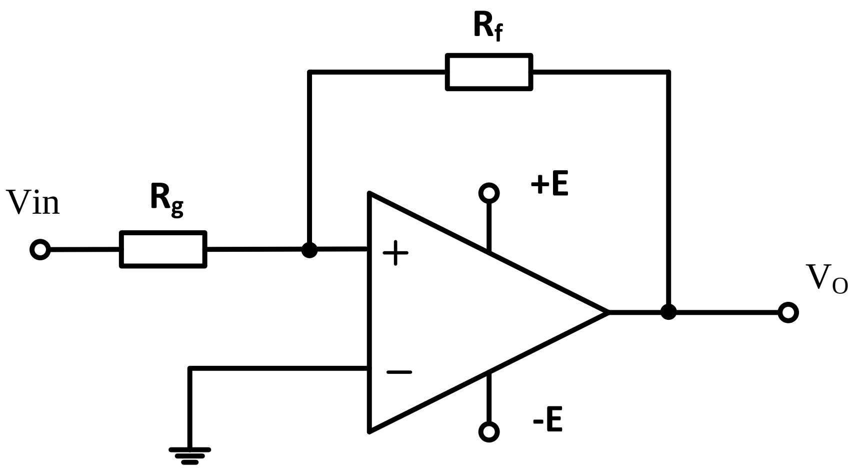


$$V_{prag} = \pm \frac{R_g}{R_f} E$$

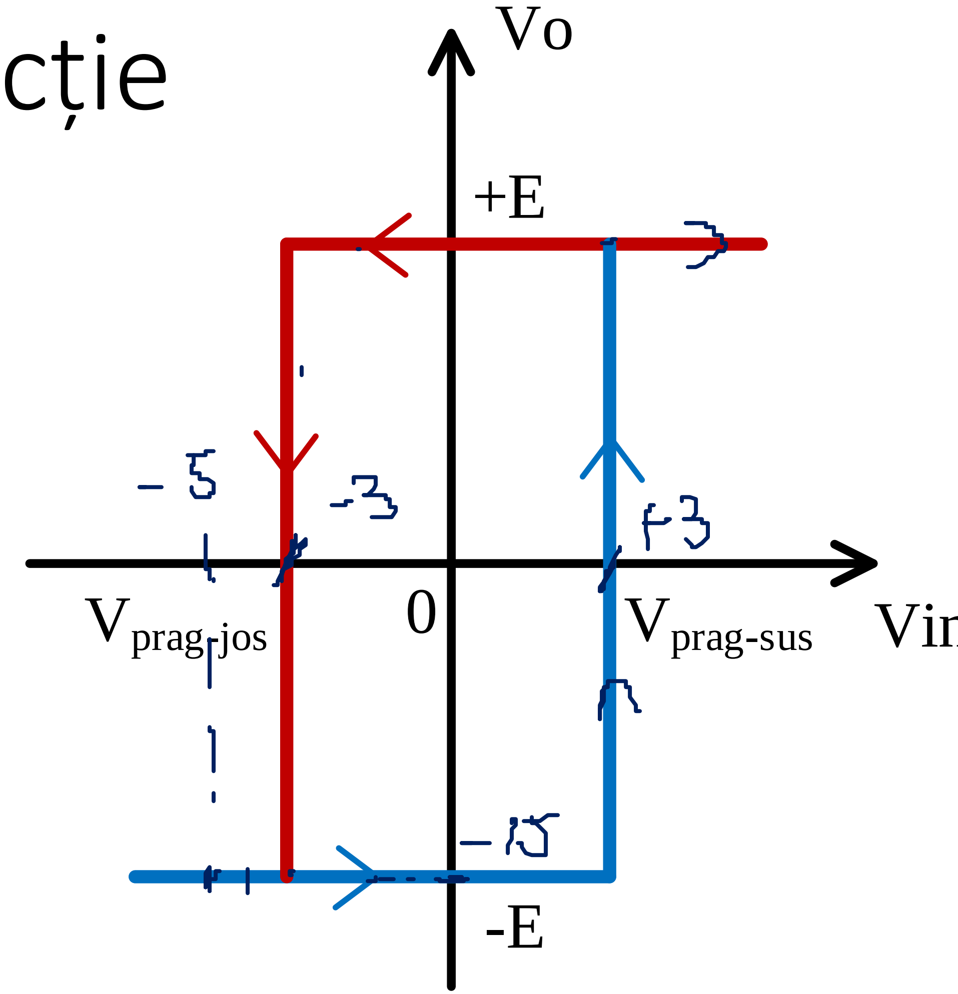


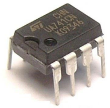


Comparator neinversor cu reacție

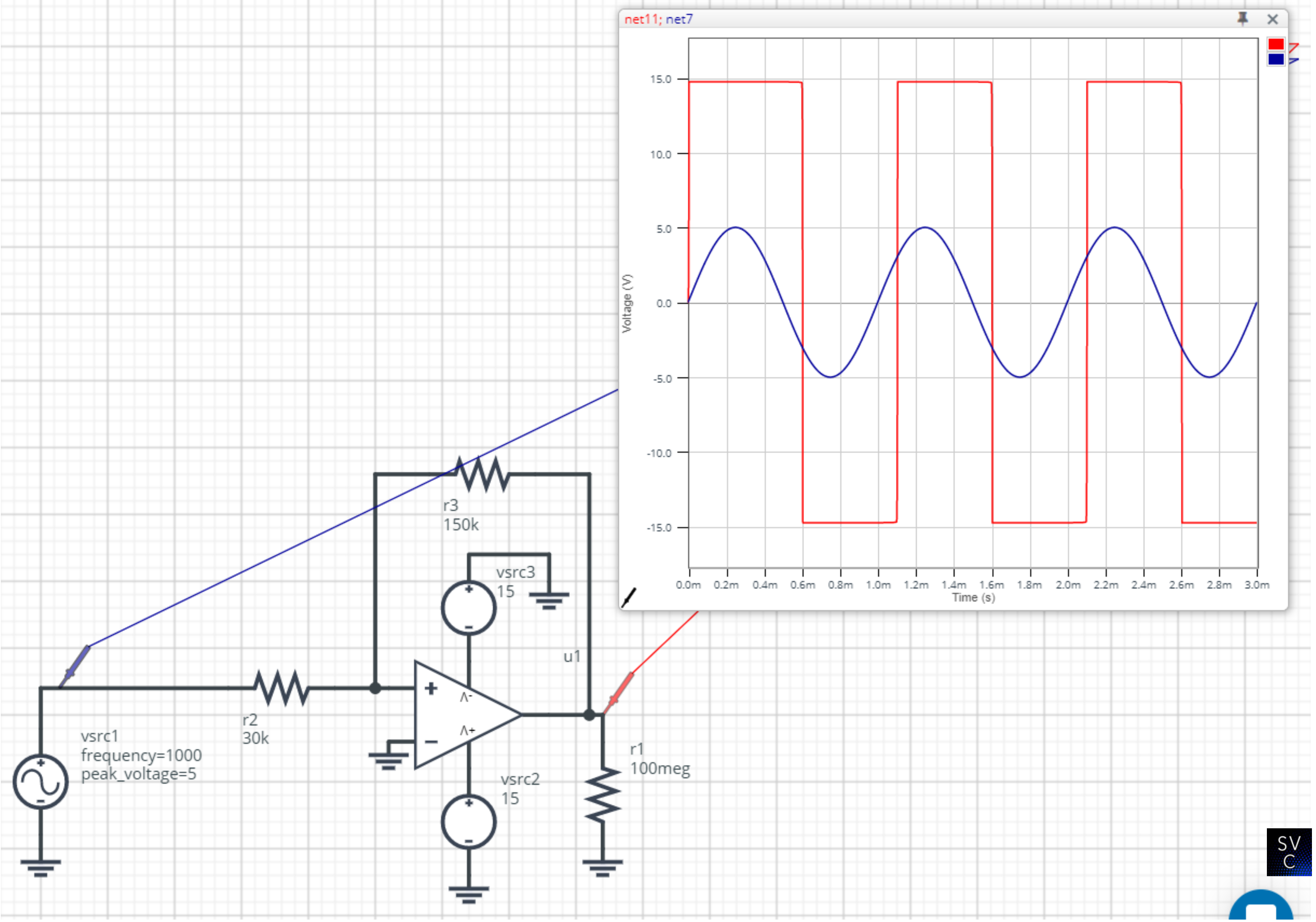


$$V_{prag} = \pm \frac{R_g}{R_f} E$$

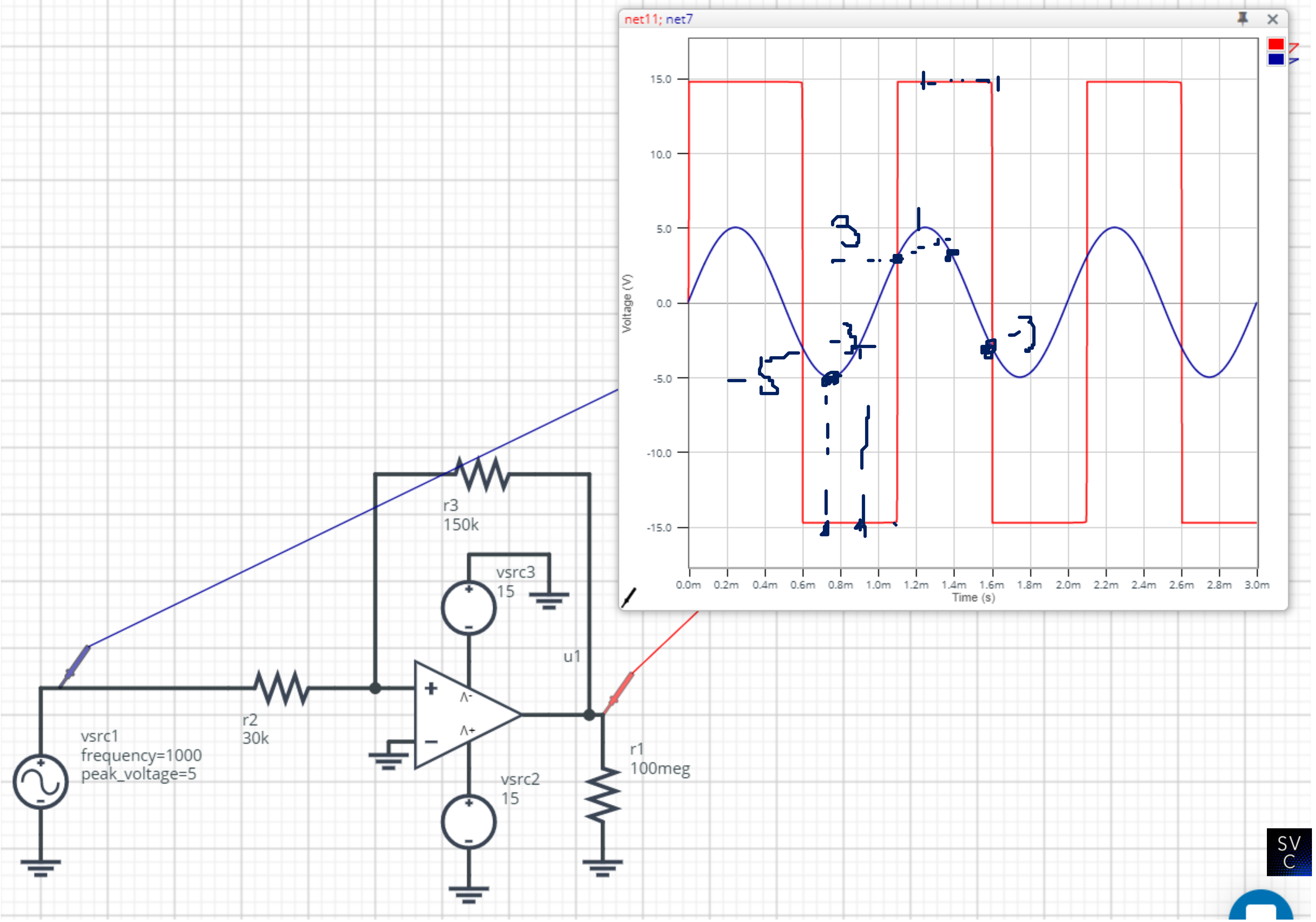




Comparator neinversor cu reacție

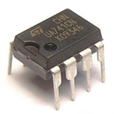


Comparator neinversor cu reacție



$$V_p = \pm \frac{30k}{150k} \times 15V$$

$$V_p = \pm 3V$$



Circuite cu AO

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  - Reacție Negativă
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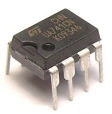
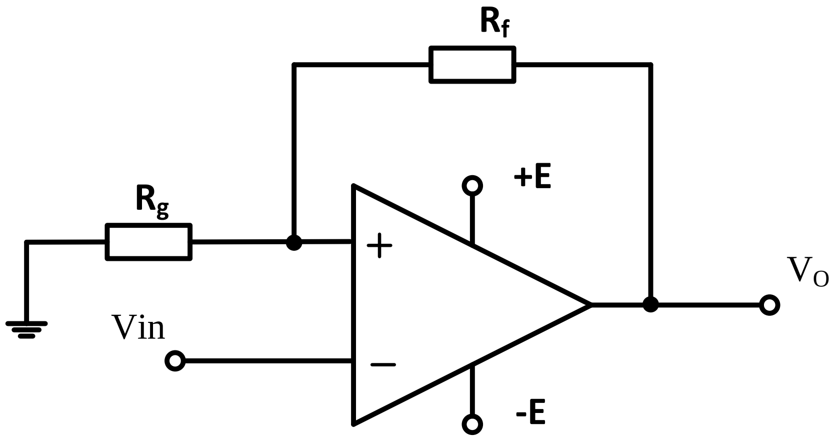
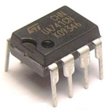
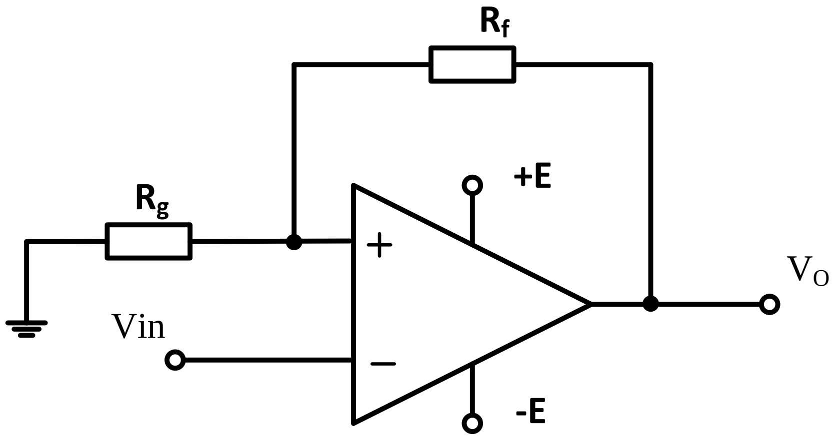


Comparator inversor cu reacție



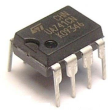


Comparator inversor cu reacție

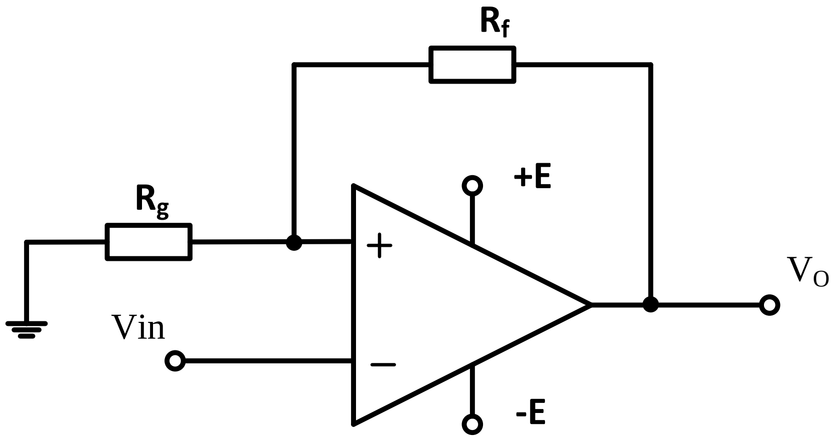


$$\left.\begin{aligned} V^- &= v_{in} \\ V^+ &= \frac{\frac{0}{R_g}+\frac{v_o}{R_f}}{\frac{1}{R_g}+\frac{1}{R_f}} \end{aligned}\right\} \Rightarrow V^+ = V^- \Rightarrow \frac{\frac{\pm E}{R_f}}{\frac{1}{R_g}+\frac{1}{R_f}} = V_{prag}$$

$$\Rightarrow \boxed{V_{prag} = \pm \frac{R_g}{R_g + R_f} E}$$

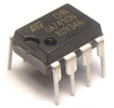


Comparator inversor cu reacție

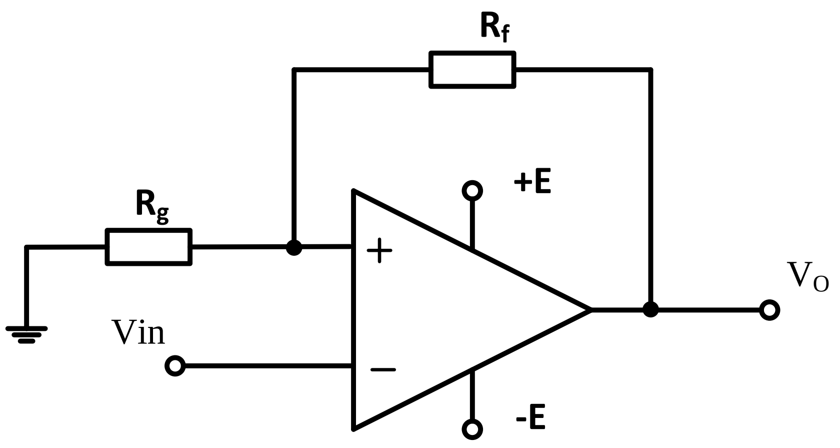


$$\left.\begin{aligned} V^- &= v_{in} \\ V^+ &= \frac{\frac{0}{R_g} + \frac{v_o}{R_f}}{\frac{1}{R_g} + \frac{1}{R_f}} \end{aligned}\right\} \Rightarrow V^+ = V^- \Rightarrow \frac{\frac{\pm E}{R_f}}{\frac{1}{R_g} + \frac{1}{R_f}} = V_{prag}$$

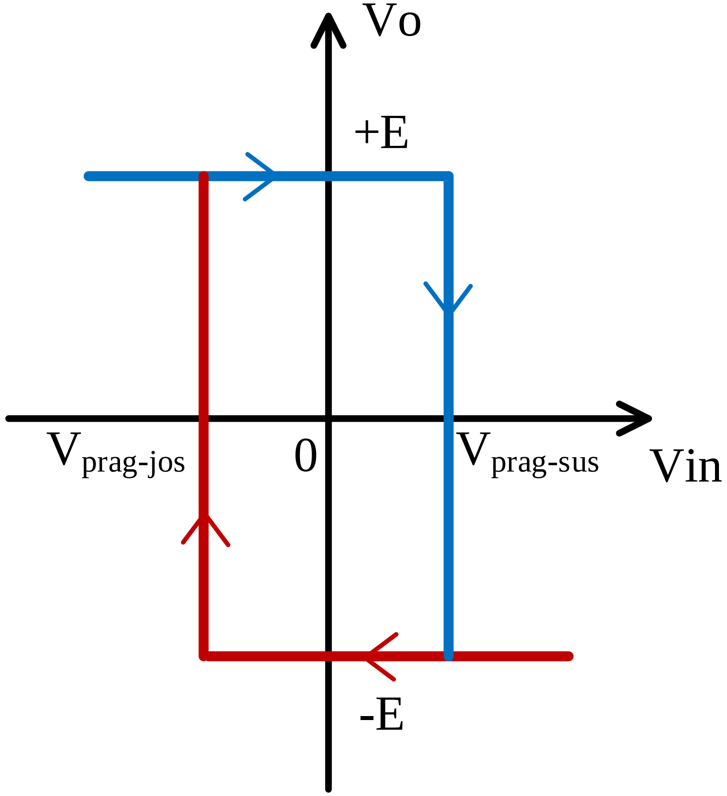
$$\Rightarrow \boxed{V_{prag} = \pm \frac{R_g}{R_g + R_f} E}$$



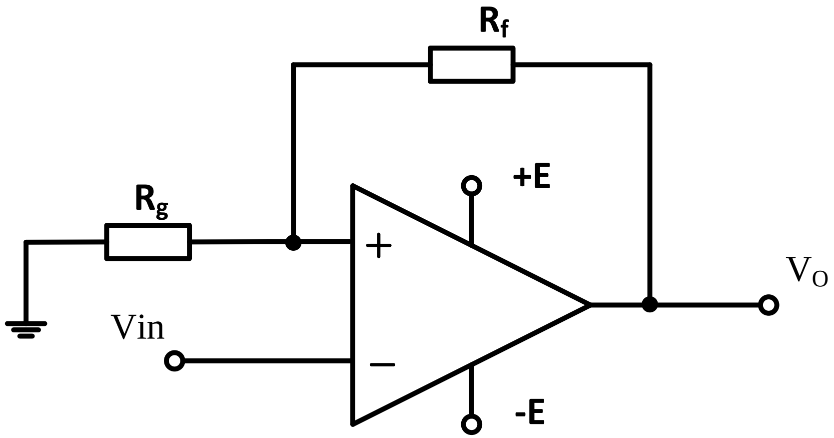
Comparator inversor cu reacție



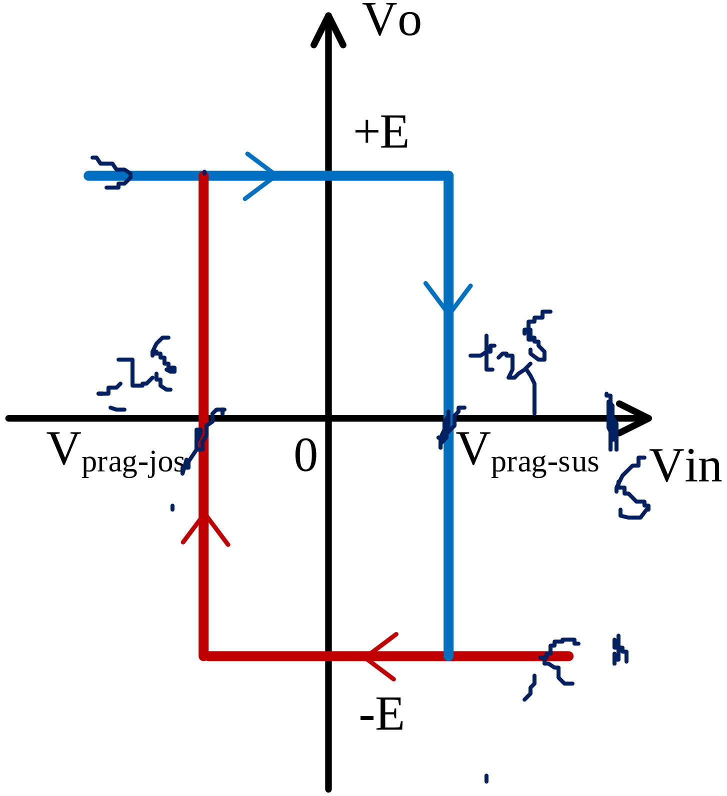
$$V_{prag} = \pm \frac{R_g}{R_g + R_f} E$$

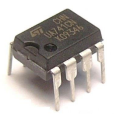


Comparator inversor cu reacție



$$V_{prag} = \pm \frac{R_g}{R_g + R_f} E$$

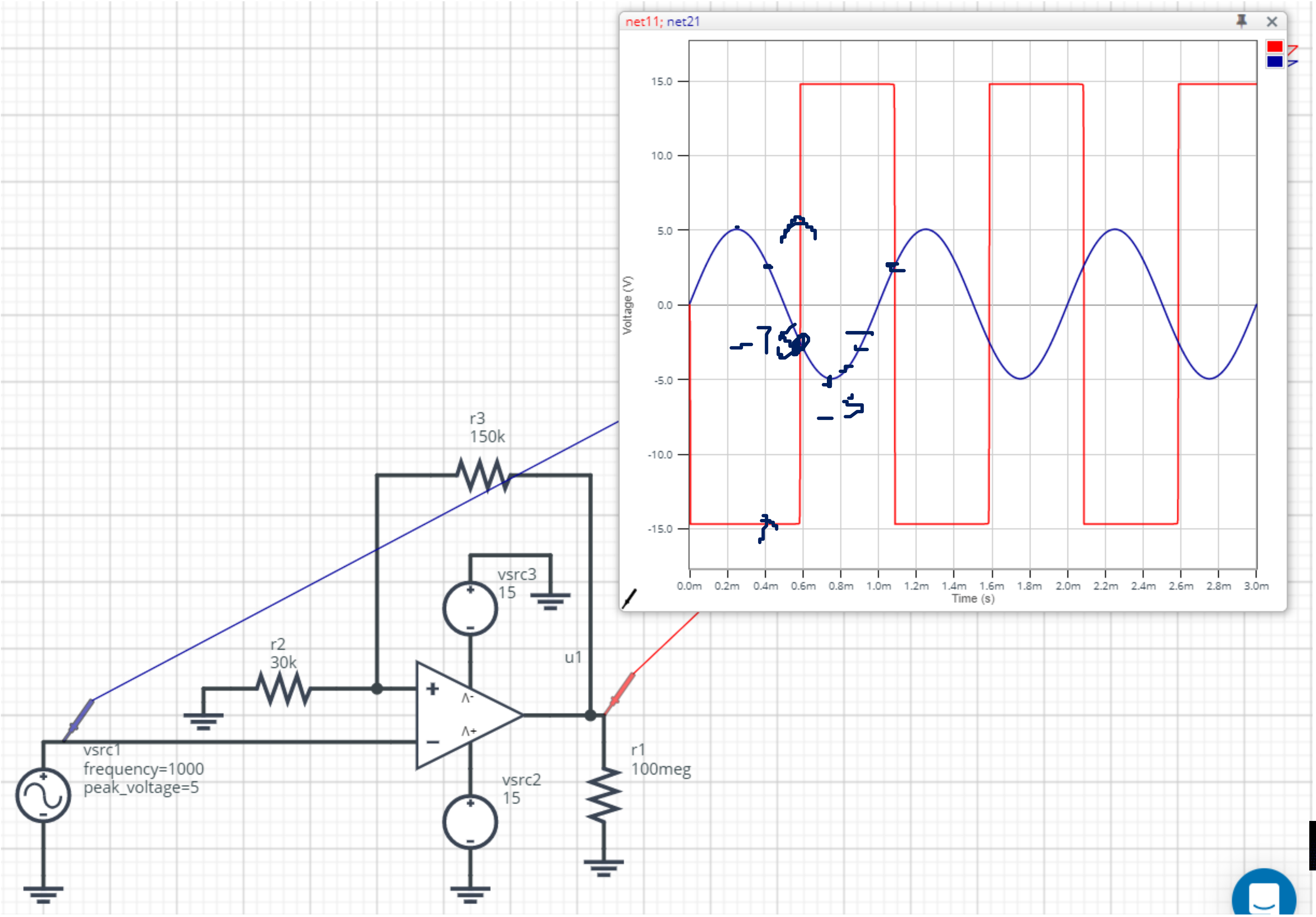


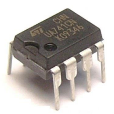


Comparator inversor cu reacție

$$V_p = \frac{30k}{30k+150k} \times 15$$

$$V_p = \pm 2{,}5V$$





Circuite cu AO

- fără Reacție
  - Comparator neinversor fără Reacție
  - Comparator inversor fără Reacție
- cu Reacție
  - Reacție Pozitivă
    - Comparator neinversor cu Reacție
    - Comparator inversor cu Reacție
  - Reacție Negativă
    - Amplificator neinversor
    - Amplificator inversor